

The Legacy of Srinivasa Ramanujan: A mathematical genius

Presenter:
J. Raveena Reddy
24R91A05H8
TKEM ,CSE-C



Srinivasa Ramanujan was a self-taught mathematician whose work profoundly influenced modern mathematics. His unique approach and **insightful theories** continue to inspire mathematicians today. This presentation will explore his remarkable journey, contributions, and the **lasting impact** he left on the world of mathematics.





Early Life and Struggles

Born in **India**, Ramanujan faced numerous challenges, including limited resources and **lack of formal education**. Despite these obstacles, his passion for mathematics led him to discover **complex concepts** that baffled many. His journey exemplifies the power of **perseverance** and **dedication** in the pursuit of knowledge.

Mathematical Contributions

Ramanujan's contributions include the **Ramanujan Prime** and **Ramanujan-Hardy number**. His work on **modular forms** and **partition functions** opened new avenues in number theory. His **innovative formulas** and **theorems** continue to be pivotal in mathematical research and application, showcasing his **genius**.

1729

The smallest number expressible
as the sum of two cubes
in two different ways.

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

Ramanujan's Partition Formula

$$p(n) = \frac{1}{2\pi\sqrt{23}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \cdot \frac{d}{dn} \left(\frac{1}{\sqrt{n - \frac{1}{24}}} \exp \left[\frac{\pi}{k} \sqrt{\frac{2}{3} \left(n - \frac{1}{24} \right)} \right] \right)$$

where

$$A_k(n) = \sum_{0 \leq m < k, (m,k)=1} e^{2\pi i (n(m,k) - 2nm/k)}$$

Srinivasa Ramanujan

1887 - 1920

Ramanujan Prime :

R_n is the smallest positive integer
such that if $\forall x \geq R_n$ then $\pi(x) - \pi\left(\frac{x}{2}\right) \geq n$

Mathematically :

$$\pi(x) - \pi\left(\frac{x}{2}\right) \geq n; \quad \forall x \geq R_n, \quad n \geq 1$$

Note : $\pi(x)$ is prime counting function

#mathematician

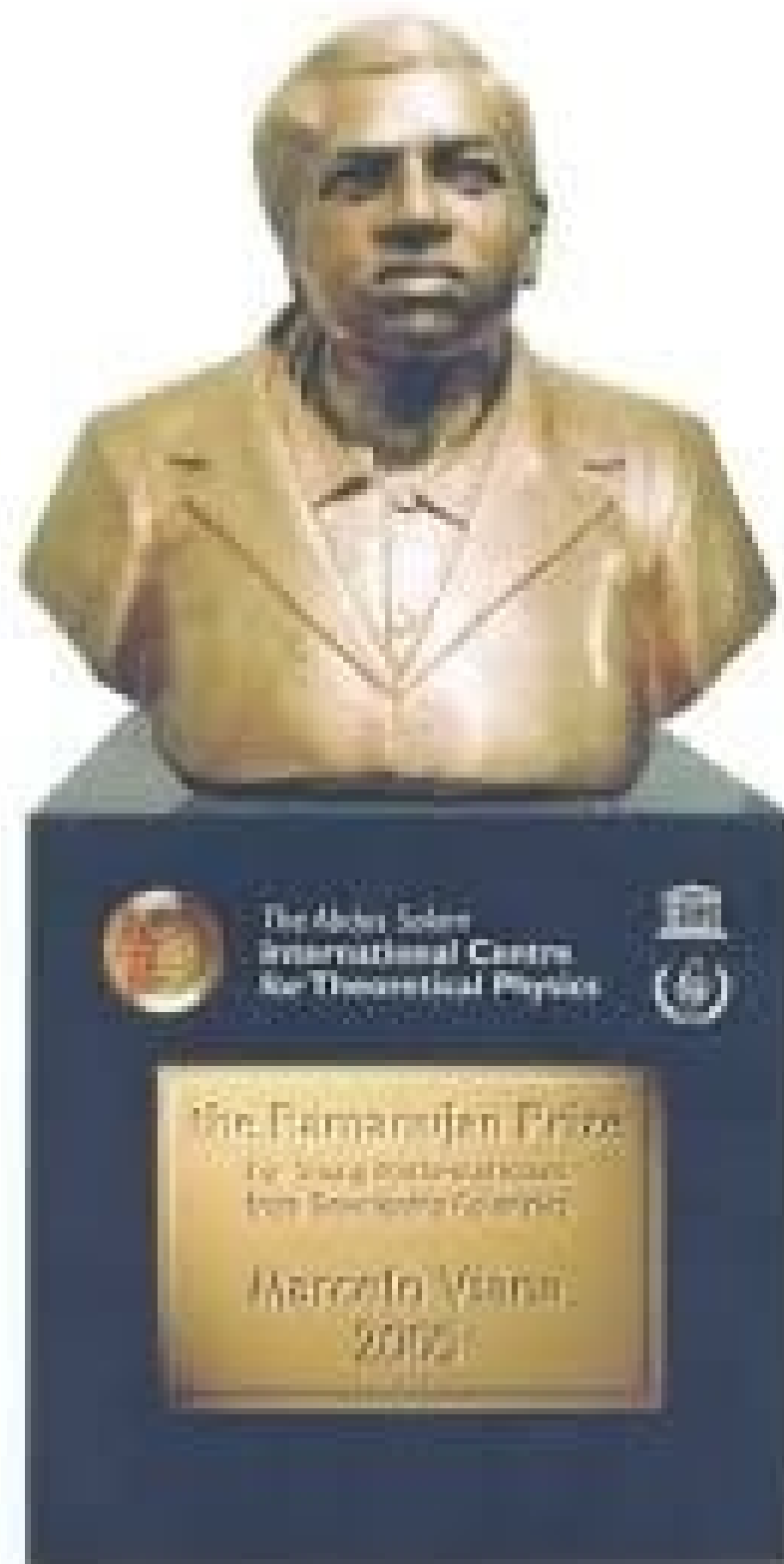
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Collaboration with Hardy

Ramanujan's collaboration with the British mathematician **G.H. Hardy** was pivotal. Hardy recognized Ramanujan's **extraordinary talent** and invited him to England, where they worked on **advanced mathematical theories**. Their partnership not only enriched mathematics but also highlighted the importance of **cultural exchange** in academia.

Ramanujan's legacy and recognition



Srinivasa ramanujan's legacy is marked by his groundbreaking contributions to number theory, infinite series, and modular forms. His work on π , partitions and famous HARDY-RAMANUJAN NUMBER (1729) continues to influence modern mathematics.

Recognition:

The Ramanujan prize was established in 2005 to honour young mathematicians from developing countries.

National mathematics day was celebrated on December 22 (Ramanujan's birthday) acknowledges his profound impact on Mathematics and education in India and globally.

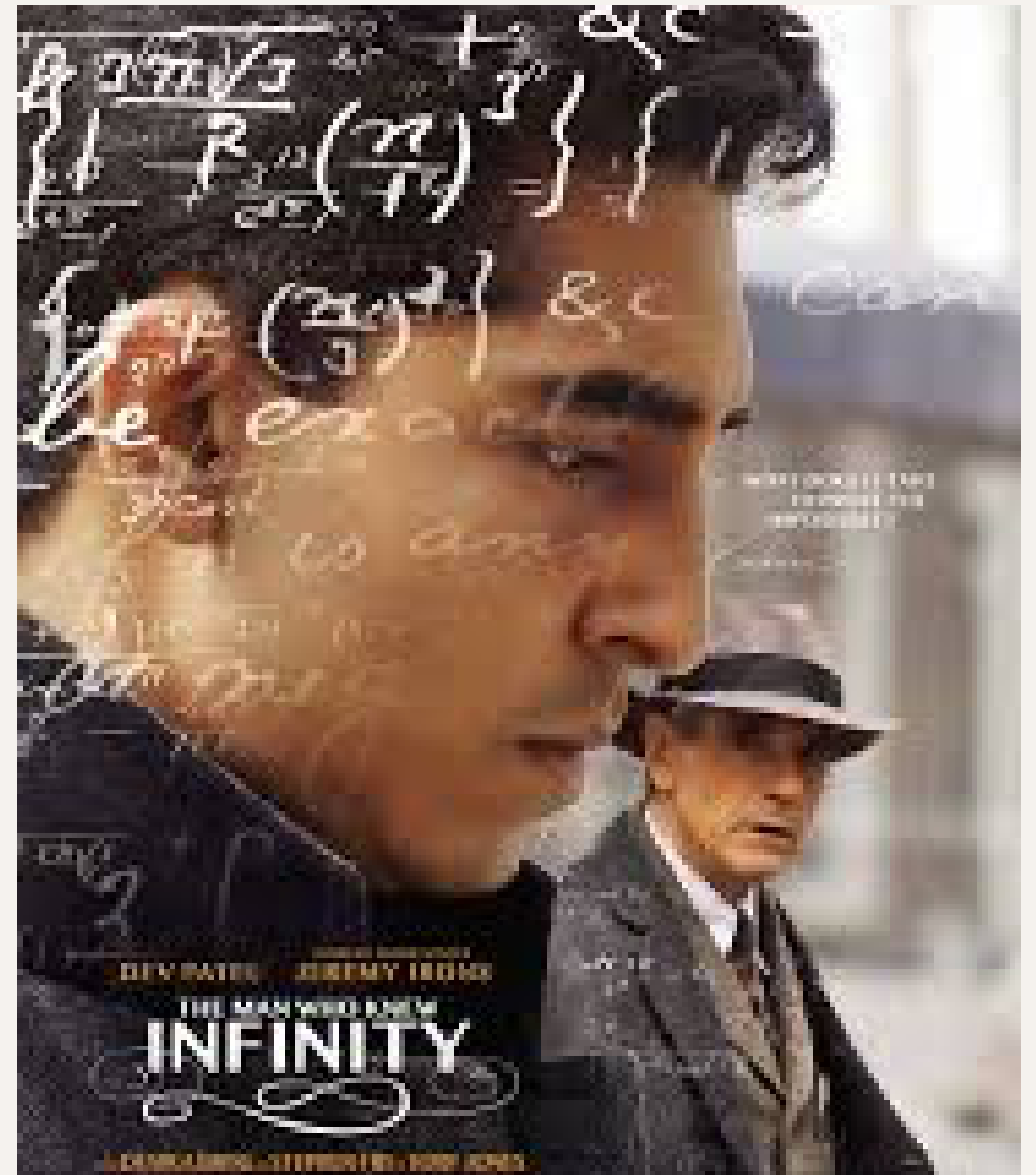
The cultural impact of Ramanujan

Ramanujan in popular culture :

References to Ramanujan in
films, literature and
documentaries (eg: the 2015
movie "the man who knew
infinity)

Inspiration for young
mathematicians :

His story serves as an inspiration
for students and professionals,
showing that passion, dedication



Conclusion

Ramanujan's legacy is celebrated globally and his work continues to inspire mathematicians.

National mathematics day (December 22) is dedicated to honouring his contributions.

